

Statistical Learning

- Prediction:** predict Y using $\hat{f}(X)$, where the exact form of $\hat{f}$ is not cared about.
- $E(Y - \hat{Y})^2 = E[f(X) + \varepsilon - \hat{f}(X)]^2 = \underbrace{[f(x) - \hat{f}(X)]^2}_{\text{reducible}} + \underbrace{Var(\varepsilon)}_{\text{irreducible}}$
- Inference:** explore how Y changes as a function of predictors X_i , not necessarily to make predictions.
- How to estimate f?
 - Parametric approach:** hypothesise a model and try to fit the parameters.
 - Non-parametric approach:** try to find a $\hat{f}$ that's close to the data points, requiring a large number of observations.
- Flexibility of a model is a trade-off between accuracy and interpretability when not overfitting.
- Bias-variance trade-off:**
 - Expected test MSE = $E[y_0 - \hat{f}(x_0)]^2 = Var(\hat{f}(x_0)) + [Bias(\hat{f}(x_0))]^2 + Var(\varepsilon)$
 - When flexibility increases, bias initially decreases faster than the variance increases. Test MSE decreases.
 - After reaching the point where bias stops decreasing very much anymore, test MSE increases.
- Classification:**
 - Bayes Classifier: $\hat{y}_0 = \operatorname{argmax}_j P(Y = j | X = x_0)$
 - Bayes Error Rate: $1 - E[\max_j P(Y = j | X)]$
 - Bayes error rate > 0 iff classes overlap (irreducible error).
- KNN: $\hat{P}(Y = j | X = x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} \mathbb{1}(y_i = j)$
- Small K: low bias. Big K: low variance.

Linear Regression

Simple Linear Regression

- Model: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$
- OLS estimates: $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$, $\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$
- The population regression line is $Y = \beta_0 + \beta_1 X$ (true parameters).
- The least squares line is $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$ (estimated parameters, changes with different training data).
- $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased estimators of β_0 and β_1 .
- $SE(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]$, $SE(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$, where $\sigma^2 = \text{Var}(\varepsilon)$.
- In practice, σ^2 is unknown and is estimated by $RSE^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2$.
- Approximate 95% confidence interval for β_j is $\hat{\beta}_j \pm 2 \times SE(\hat{\beta}_j)$.
- t-test: for $H_0 : \beta_j = 0$ and $H_1 : \beta_j \neq 0$, calculate $t = \frac{\hat{\beta}_j - 0}{SE(\hat{\beta}_j)}$, where $t \sim t_{n-2}$.
- Accessing model accuracy:**
 - RSE measures the lack of fit of the model to the data.

- R-squared: proportion of variance explained by the model. $R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$, where $TSS = \sum_{i=1}^n (y_i - \bar{y})^2$ and $RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$.
- For simple linear regression, $R^2 = r^2$, where $r = \hat{\text{Cor}}(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$.

Multiple Linear Regression

- Model: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \dots + \hat{\beta}_p x_{ip}$
- β_i is the average effect on Y of one unit increase in X_i , **holding all other predictors fixed**.
- F-test:**
 - $H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0$ and $H_1 : \text{at least one } \neq 0$, calculate $F = \frac{(TSS - RSS)/p}{RSS/(n - p - 1)}$, where $F \sim F_{p, n - p - 1}$.
 - Under H_0 , F is around 1. Under H_1 , F is greater than 1.
 - Even if H_0 is true, individual p-values for t-test can be significant due to inflation of Type I error in multiple testing.
- Model selection:**
 - benchmark: RSS
 - stopping rule: p-value
 - forward, backward selection: $1 + \frac{p(p+1)}{2}$ models, mixed selection
- Model accuracy:** $RSE = \sqrt{\frac{1}{n - p - 1} RSS}$, R^2 and graph (to investigate synergy).
- Prediction interval** accounts for both the reducible and irreducible error, hence targets individual observations.
- Confidence interval** only accounts for the reducible error, hence targets the average and is narrower.

Other Considerations

- For predictors with $k \geq 2$ levels: use $k - 1$ dummy variables.
- Interaction term** coefficient interpretation: the increase in the effectiveness of A for a one-unit increase in B.
- If the interaction term is significant, the main effects should be kept in the model.
- Interaction term** of a qualitative with a quantitative variable gives different slopes for different levels of the qualitative variable.
- Non-linear relationships:** polynomial regression.

Problems and Solutions

- Non-linearity: identify using residual plots (e against x/y), add powers of x, log(x), $\sqrt{x}$, etc.
- Correlation in ε : makes SEs, CIs and PIs **underestimated**. Identify with residual plots.
- Non-constant variance (heteroscedasticity): identify with residual plots, transform with $\sqrt{Y}$, $\log Y$, etc. When the variance follows a function, use weighted least squares with weights inversely proportional to the variance.
- Outliers: identify with studentized residuals (typically > 3).

- High leverage points: calculate $h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^n (x_j - \bar{x})^2}$. Average leverage is $\frac{p+1}{n}$.
- Collinearity: $\hat{\beta}$ for correlated predictors have high variance, hence large SEs and p-values. $VIF(\hat{\beta}_j) = \frac{1}{1 - R_{X_j | X_{-j}}^2}$, where $R_{X_j | X_{-j}}^2$ is the R^2 of X_j regressed on all other predictors. $VIF > 5$ is a sign of collinearity.

KNN Regression

- $\hat{f}(x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} y_i$.
- When p = 1, KNN is much better than OLS for non-linear relationships.
- For higher ps, KNN is not as good as OLS: curse of dimensionality.

Classification

Logistic Regression

- $p(X) = P(Y = 1 | X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$
- logit: $\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X$
- $L(\beta_0, \beta_1) = \prod_{i: y_i = 1} p(x_i) \prod_{i: y_i = 0} (1 - p(x_i))$

Linear Discriminant Analysis

- $\pi_k = P(Y = k)$, $f_k(X) = P(X = x | Y = k)$
- $p_k(x) = P(Y = k | x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}$, posterier probability.
- Assume $f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp(-\frac{1}{2\sigma_k^2}(x - \mu_k)^2)$, and $\sigma_k = \sigma$.
 $p_k(x) = \frac{\pi_k \exp(-\frac{1}{2\sigma^2}(x - \mu_k)^2)}{\sum_{l=1}^K \pi_l \exp(-\frac{1}{2\sigma^2}(x - \mu_l)^2)}$.
- $\delta_k(x) = x \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$
- Estimates: $\hat{\pi}_k = \frac{n_k}{n}$, $\hat{\mu}_k = \frac{1}{n_k} \sum_{i: y_i = k} x_i$, $\hat{\sigma}^2 = \frac{1}{n - K} \sum_{k=1}^K \sum_{i: y_i = k} (x_i - \hat{\mu}_k)^2$
- For $p > 1$, $\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log(\pi_k)$, where Σ is the covariance matrix.
- Sensitivity:** true positive rate, **specificity:** true negative rate. Nudge the softmax threshold to adjust sensitivity and specificity.
- ROC curve: plot sensitivity against 1 - specificity.

Quadratic Discriminant Analysis

$$\delta_k(x) = -\frac{1}{2} \log |\Sigma_k| - \frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) + \log(\pi_k)$$
$$= -\frac{1}{2} x^T \Sigma_k^{-1} x + x^T \Sigma_k^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma_k^{-1} \mu_k - \frac{1}{2} \log |\Sigma_k| + \log(\pi_k)$$

- $p(p + 1)/2$ parameters for Σ_k .

Resampling Methods

- Validation set: variable MSE, tends to overestimate test error rate.

- LOOCV: far less bias but costly. **For least squares**, $CV(n) = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{1 - h_i} \right)^2$, where h_i is the leverage.
- LOOCV vs. k-fold CV: highly positively correlated MSEs result in higher variance. More pronounced for big p and small n.
- Bootstrap:
 - Minimise $\text{Var}(\alpha X + (1 - \alpha)Y)$: $\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$
 - Bootstrap resampling: sample with replacement n observations, repeat B times.
 - If bootstrap SE estimates for regression models differ from the analytical SE estimates, the model is not a good fit.

Linear Model Selection and Regularization

Subset Selection

Best subset selection

- Use RSS to compare among models with the same k, then CV to compare across the p + 1 models. 2^p models.

Stepwise selection

- Reduce the search space because of computational limitations and avoid finding "too good" models that overfit.
- In each step, extends the model by adding or removing one predictor with the smallest RSS or largest R^2 .
- Among the p + 1 models obtained, choose using CV, C_p , AIC, BIC, adjusted R^2 .
- $1 + p(p + 1)/2$ models. However, when $n < p$, only have $\mathcal{M}_0$ up to $\mathcal{M}_{n-1}$.

Model selection criteria

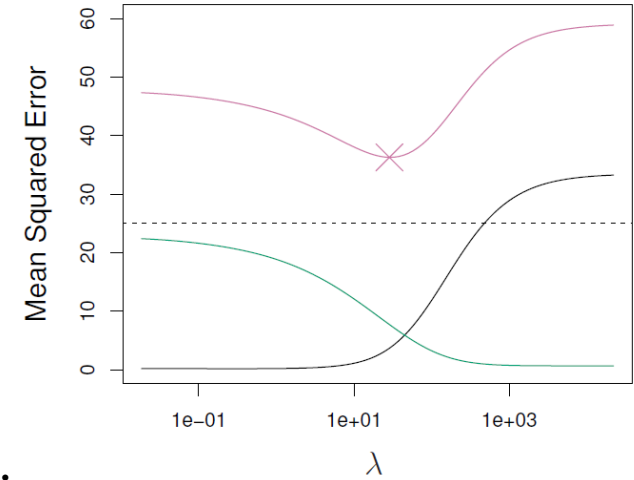
- $C_p = \frac{1}{n}(RSS + 2d\hat{\sigma}^2)$, where d is the number of predictors and $\hat{\sigma}^2$ is the estimated $\text{Var}(\varepsilon)$. Typically, $\hat{\sigma}^2$ is estimated using the full model. Theory: C_p is an unbiased estimate of test MSE if $\hat{\sigma}^2$ is unbiased.
- $AIC = \frac{1}{n\hat{\sigma}^2}(RSS + 2d\hat{\sigma}^2)$.
- $BIC = \frac{1}{n\hat{\sigma}^2}(RSS + \log(n)d\hat{\sigma}^2)$, heavier penalty on d.
- Adjusted $R^2 = 1 - \frac{RSS/(n-d-1)}{TSS/(n-1)}$.
- CV methods make fewer assumptions about the true model. Better when hard to: find effective number of predictors, estimate $\hat{\sigma}^2$ or when asymptotics are not realistic, e.g. n not much bigger than p.

Shrinkage Methods

Ridge Regression

- Target: $\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p \beta_j^2$.
- $X_j \hat{\beta}_{j,\lambda}^R$ depends on both λ and the scaling of X , unlike in OLS where $X_j \hat{\beta}_j$ is constant.
- Hence, standardisation is necessary to make X_i have mean 0 and sd 1.

- Ridge regression trades off a small increase in bias for a large decrease in variance. It works well for $p \approx n$ and $p > n$.



Lasso

- Target: $\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p |\beta_j|$.
- Optimisation dual: $\arg \min_{\beta} \text{RSS s.t. } \sum_{j=1}^p |\beta_j| \leq s$.
- Best subset selection: $\sum_{j=1}^p \mathbf{1}(\beta_j \neq 0) \leq s$.
- Ridge performs better when all coeffs are of roughly equal size.

Dimension Reduction Methods

- Let $Z_m = \sum_{j=1}^p \phi_{jm} X_j$, for $m = 1, 2, \dots, M$. This constrains $\hat{\beta}_j$ to be linear combinations.
- This increases the bias, but reduces the variance if p is large and $M \ll p$.
- If $M=p$ and Z_m are orthogonal, no change.

Principal Components Regression

- First principle component: $Z_1 = \phi_{11}(X_1 - \bar{X}_1) + \phi_{21}(X_2 - \bar{X}_2) + \dots + \phi_{p1}(X_p - \bar{X}_p)$, where $\text{Var}(Z_1)$ is maximised subject to $\sum_{j=1}^p \phi_{j1}^2 = 1$.
- Assumption: the directions of Z_i s are associated with the response.
- Ridge regression is closely related to PCR.
- Favours bigger scale predictors, hence need to standardise.

Partial Least Squares

- First PLS direction: Standardise all X_i s. ϕ_{j1} = coefficient of $\text{SLR}(Y \sim X_j)$
- Second: For $k = 1, \dots, p$, run $\text{SLR}(X_k \sim Z_1)$ to obtain $e_k^{(1)}$ used as $X_k^{(1)}$.

High Dimensional Problems

- When p is large, the multicollinearity problem is severe. The set of signal predictors obtained from one dataset may not be the true predictors.
- Test error on an independent test set or CV errors are the only good measures.

Unsupervised Learning

PCA

- Computation: Center $\mathbf{X}$. Solve $\max \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^p \phi_{j1} x_{ij} \right)^2$ subject to $\sum_{j=1}^p \phi_{j1}^2 = 1$.
- Alternative interpretation: low-dimensional linear surfaces that are closest to the observations.
- When $M = \min(n - 1, p)$, the surface is exact.
- Need to standardise.
- Proportion of variance explained (PVE) of the m th principle component: $\frac{\sum_{i=1}^n \left(\sum_{j=1}^p \phi_{jm} x_{ij} \right)^2}{\sum_{i=1}^n \sum_{j=1}^p x_{ij}^2}$.

Clustering

K-means

- Minimise pairwise squared Euclidean distance: $\frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2$.
- Local minimum algorithm: 1. Randomly assign clusters. 2. Compute centroids. 3. Assign each observation to the closest centroid.

Hierarchical Clustering

- Caveat: The true structure may not be nested. (e.g. gender and nationality)
- Linkage: complete (max pairwise dissimilarity), single (min), average (mean), centroid (between centroids).
- Average and complete preferred, single leads to one-at-a-time fusion.
- Correlation-based distance: focuses on the shapes of the observations rather than the magnitudes.
- Standardisation may be necessary.

Remarks

- Outliers or perturbations can have a big impact.

R

- `lm()` summary reports t-statistics and p-values for each predictor, **after adjusting for other predictors**.
- `hatvalues()` returns the leverage of each observation.
- `anova()` compares models.
- `glmnet()` alpha = 0 for ridge, alpha = 1 for lasso.
- `pcr()` $CV = \sqrt{MSE}$
- square sdev and divide by total var to get PVE.